Microwave Oven



Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Microwave; Microwave Oven

What it is used for

A microwave oven heats or cooks food using microwave energy inside a closed cooking cavity. Use microwave-safe containers, follow the food instructions, and check that food reaches a safe temperature before serving.

Main parts

- Cooking cavity and door
- Door seal, hinges, and safety interlocks
- Control panel, timer, and power settings
- Turntable and roller ring, if fitted
- Magnetron and internal electrical components
- Power cord and plug

Before you start

1. Place the oven on a stable surface with the required ventilation clearance.
2. Inspect the door, hinges, seal, turntable, cord, and plug for damage.
3. Confirm that the oven cavity is clean and free of metal objects.
4. Use only containers labelled safe for microwave cooking.
5. Check food packaging and follow the recommended cooking or reheating instructions.
6. Ensure the door closes fully and the safety interlocks are not damaged.

How to use

1. Place the food in a suitable microwave-safe container.
2. Cover the food with a microwave-safe lid or cover that allows steam to escape.
3. Position the container in the centre of the turntable or cooking area.
4. Close the door securely.
5. Select the required power level and cooking time.
6. Start the oven and monitor operation without opening the door unnecessarily.
7. Pause and stir or rotate food when the instructions recommend it.
8. Use suitable heat protection when removing hot containers.
9. Check that the food is heated evenly and reaches the recommended safe temperature.
10. Allow standing time when required by the recipe or package instructions.
11. Switch off the oven and remove the food carefully.
12. Wipe up spills after the cavity has cooled sufficiently.
13. Switch off and unplug the oven before cleaning or when required by the instructions.

⚠ Safety

- Never operate the oven with a damaged door, seal, or safety interlock.
- Do not operate the oven when empty unless the manufacturer specifically permits it.
- Do not place metal utensils, foil, or unsuitable containers inside the oven.
- Open sealed containers and pierce suitable food skins as instructed to prevent pressure build-up.
- Beware of steam when uncovering food and liquids, which may become superheated.
- Never remove the outer casing or attempt magnetron repairs; servicing must be done by qualified personnel.

Care & storage

- Clean the cavity, door seal, and turntable regularly using a suitable cleaner.
- Keep the door seal and interlock areas free from food debris.
- Avoid abrasive tools that may damage the interior or door surface.
- Inspect the power cord and door operation regularly.
- Arrange qualified service if the oven sparks unexpectedly, overheats, or the door fails to close properly.

Common mistakes to avoid

- Using metal or non-microwave-safe containers.
- Operating the oven with a damaged door seal or interlock.
- Heating food without checking for hot spots or required standing time.
- Attempting to repair internal high-voltage components without qualification.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Prepared by - Subodh Kumar